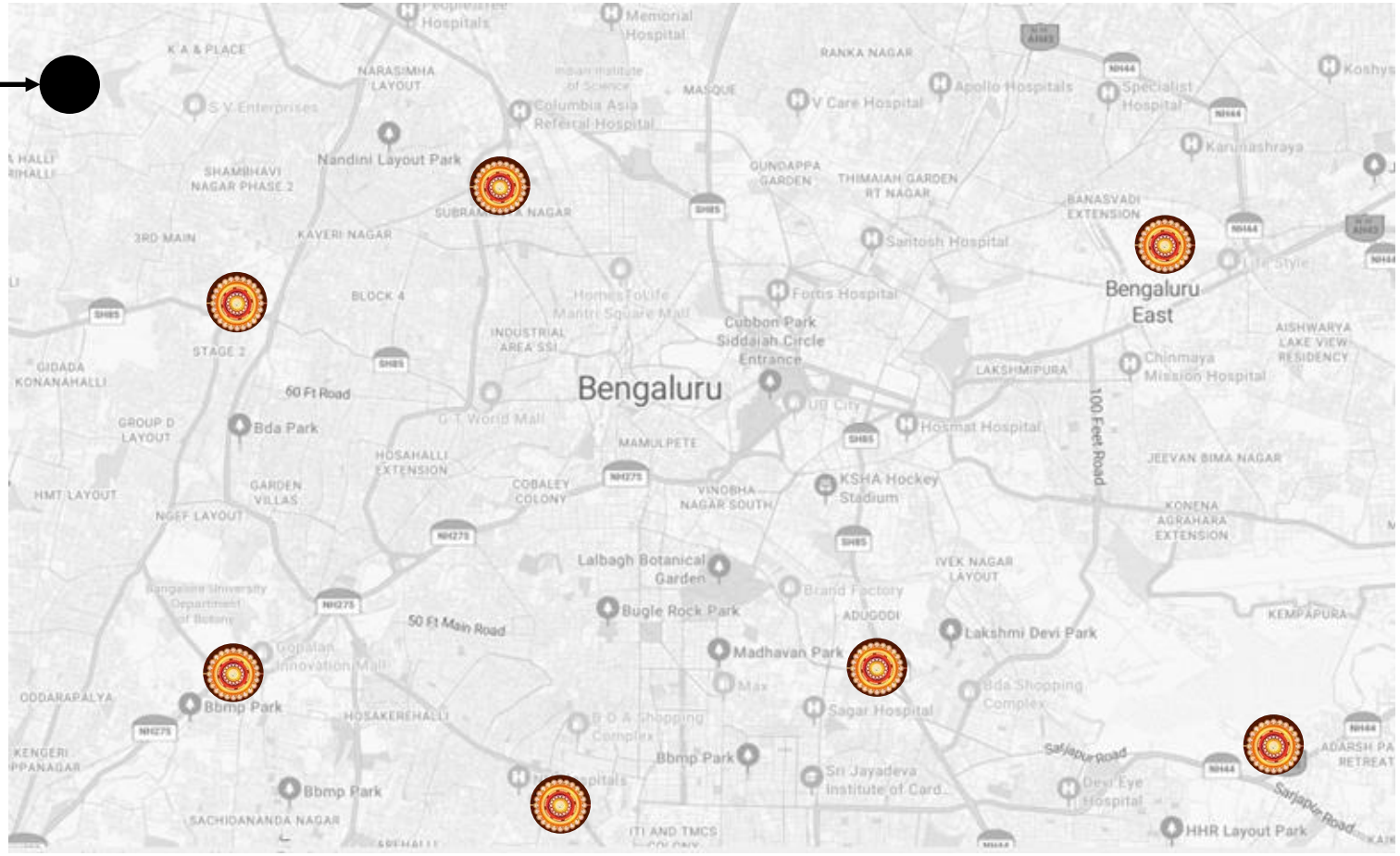
BRAVING BANGALORE TRAFFIC

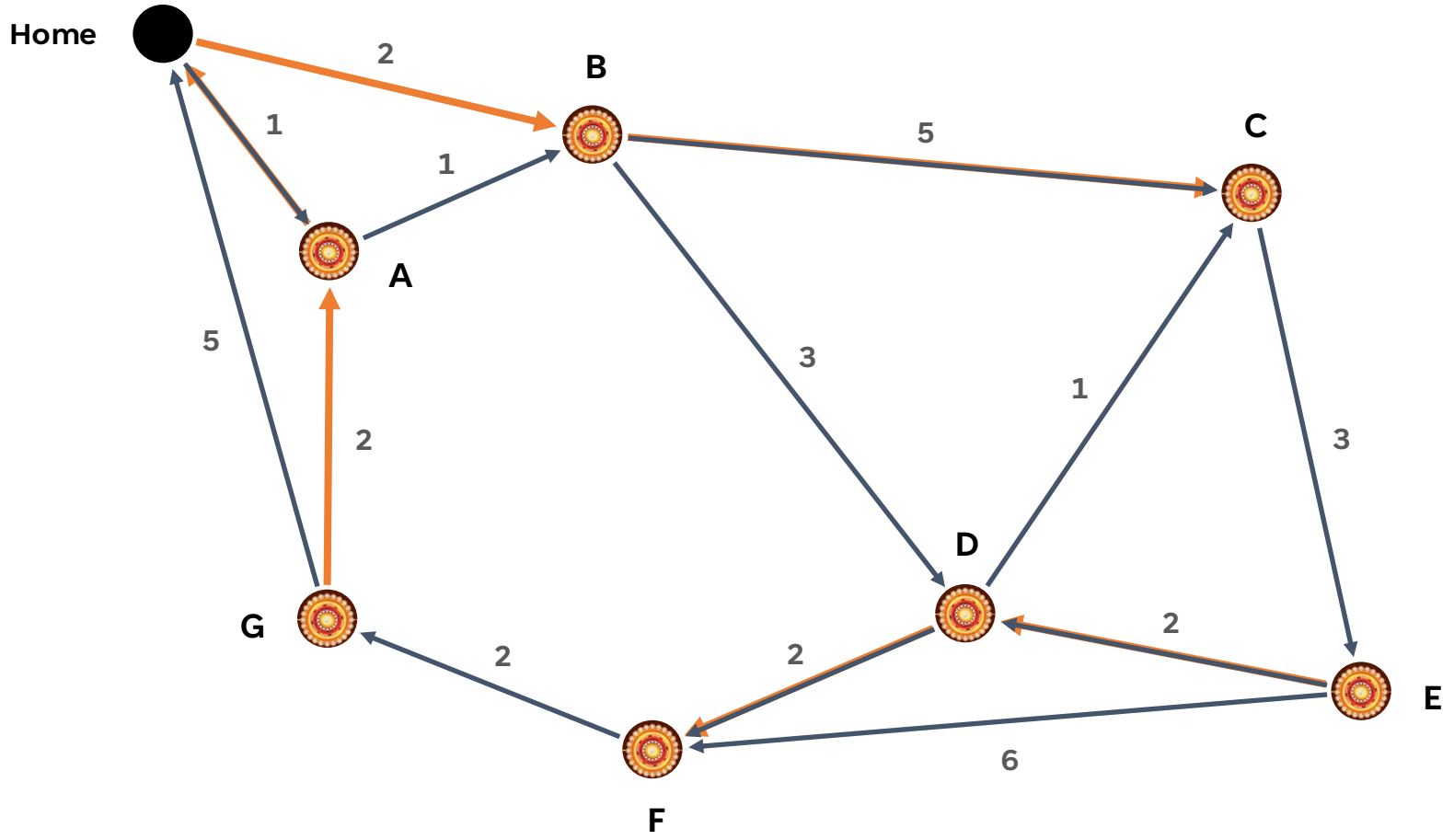
- Say it is Rakhi today, and you need to visit your 7 cousins.
- What is the shortest route that visits all cousins and returns home?

You are here. →



- In OR parlance, this is known as “Traveling Salesman Problem” (TSP).

BRAVING BANGALORE TRAFFIC: LET'S OPTIMISE IT.



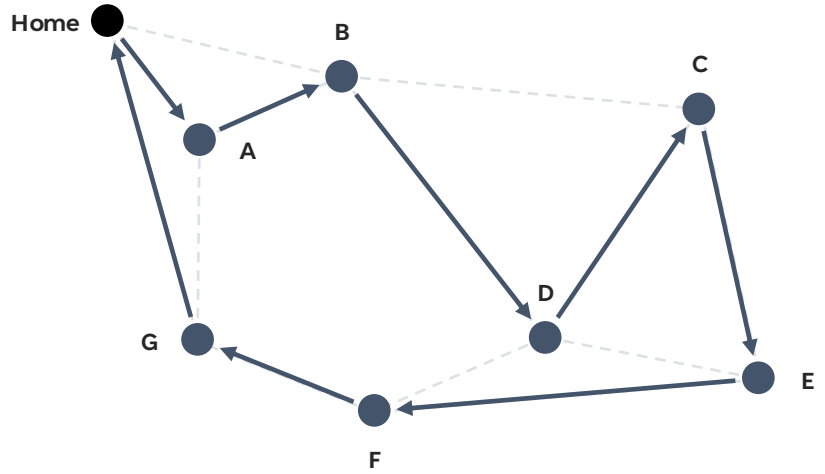
Solution using “visit the nearest cousin next”

Total distance: **19 km**

TWO KEY STEPS: BUILD, THEN IMPROVE

Construction: build a solution

Add one decision at a time by a simple rule; never look back.



Nearest cousin next: **22 km**

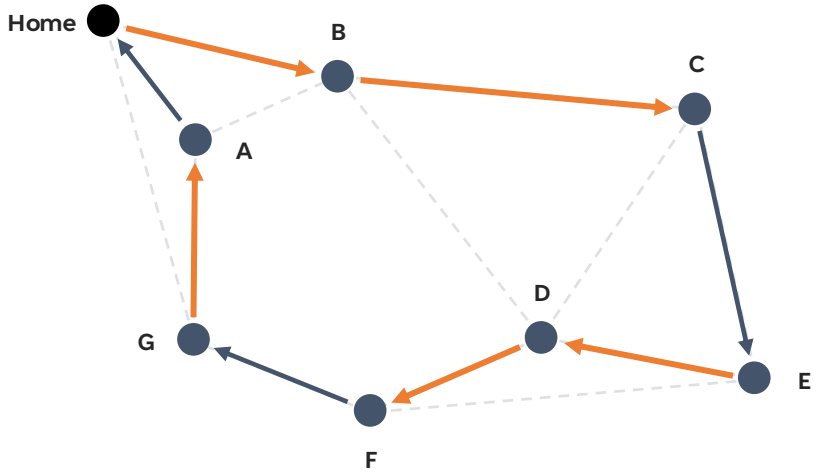
Example: greedy loot for the heist bag; sweep for vehicle routing (coming up).

Fast and simple, but rarely the best.

starting solution

Improvement: make it better

Tweak a complete solution; keep changes that help.



22 → 21 → **19 km**

Example (in next slides): neighbourhood search, 2-opt, tabu search, genetic algorithms.

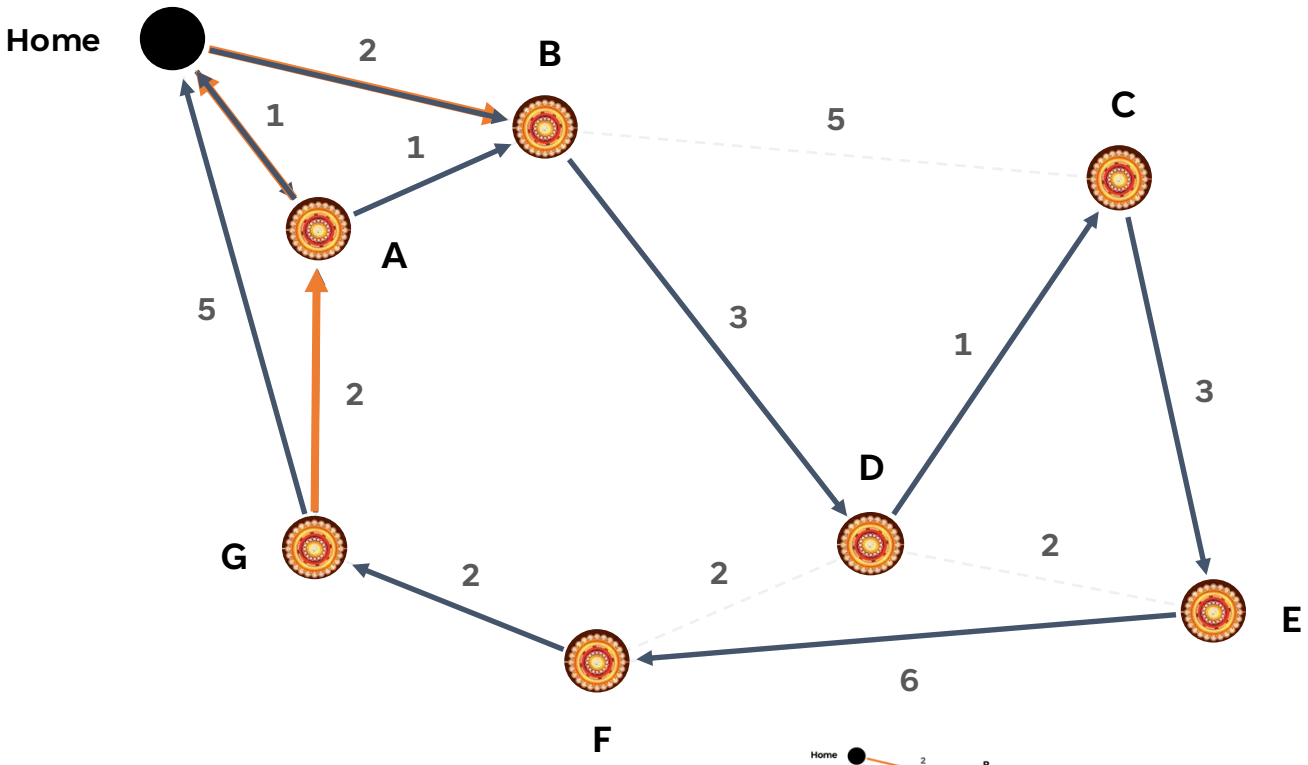
Slower, but usually much better. Can get stuck.

Practice: Construct a decent solution fast, then improve it.

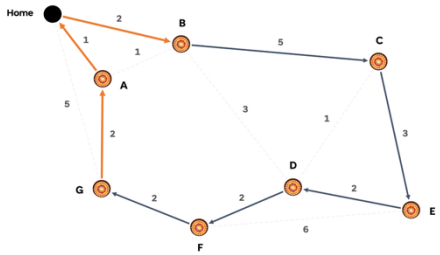


LOCAL AND GLOBAL OPTIMALITY

TSP ROUTE: STUCK IN A LOCAL MINIMUM



Route length: **20 km** stuck



Target. 19 Km

Move: swap any two stops in the visiting order. Home counts as a stop too.

Accept a move only if the route gets shorter.

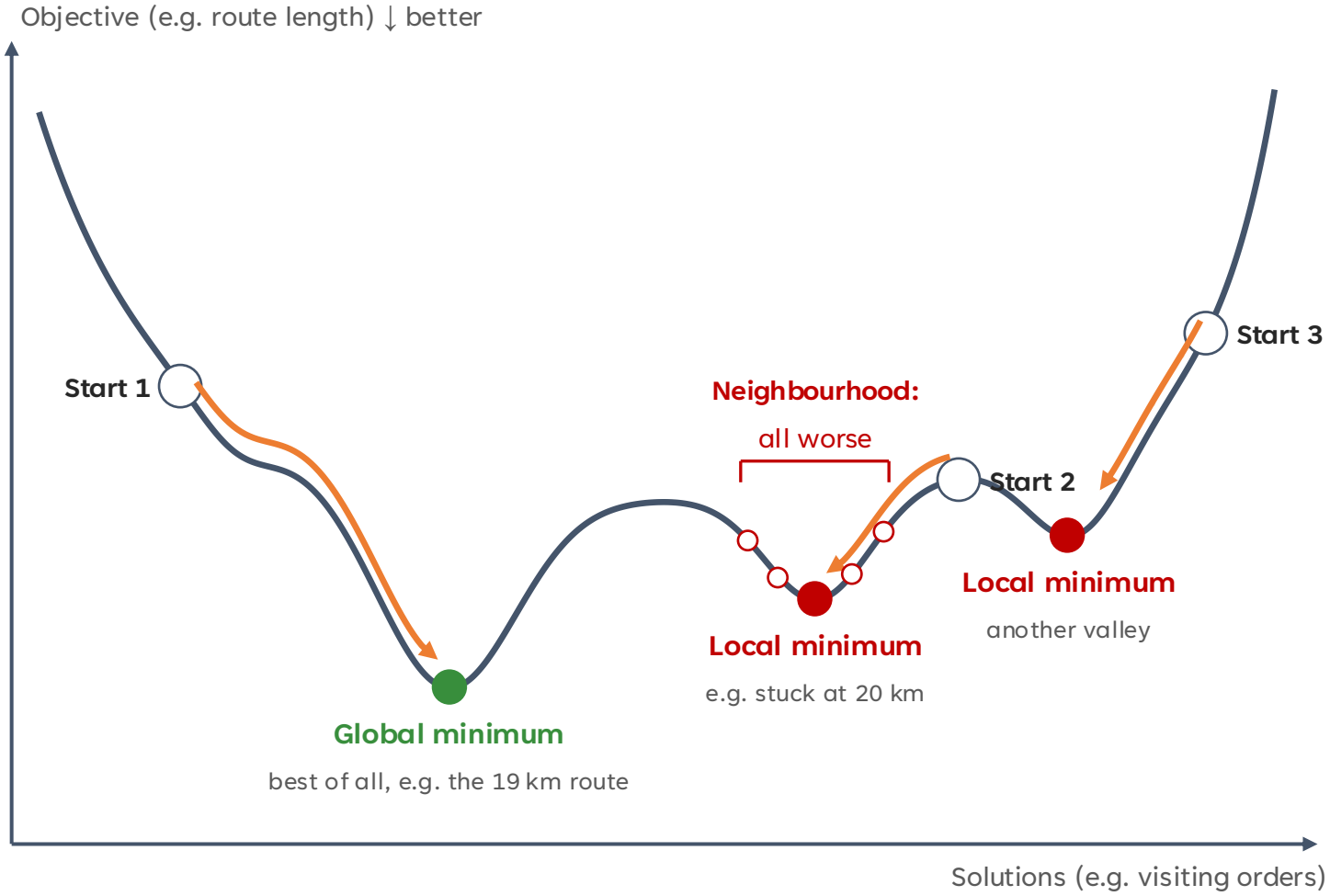
Step	Move	Route	km
Start	Nearest-cousin route	H-A-B-D-C-E-F-G-H	**22**
Iter 1	Swap Home and A	H-B-D-C-E-F-G-A-H	**20**
Iter 2	Swap C and D	H-B-C-D-E-F-G-A-H	**21** ✗
	Undo Home–A swap	H-A-B-D-C-E-F-G-H	**22** ✗

The other 26 swaps need a road that does not exist.

Every neighbour of the 20 km route is worse, so the search stops.

This is a **local minimum**. The best route is 19 km.

LOCAL AND GLOBAL MINIMUM



What hill climbing does

Walks downhill until no neighbour is better, then stops.

Where you start decides which valley you end in.

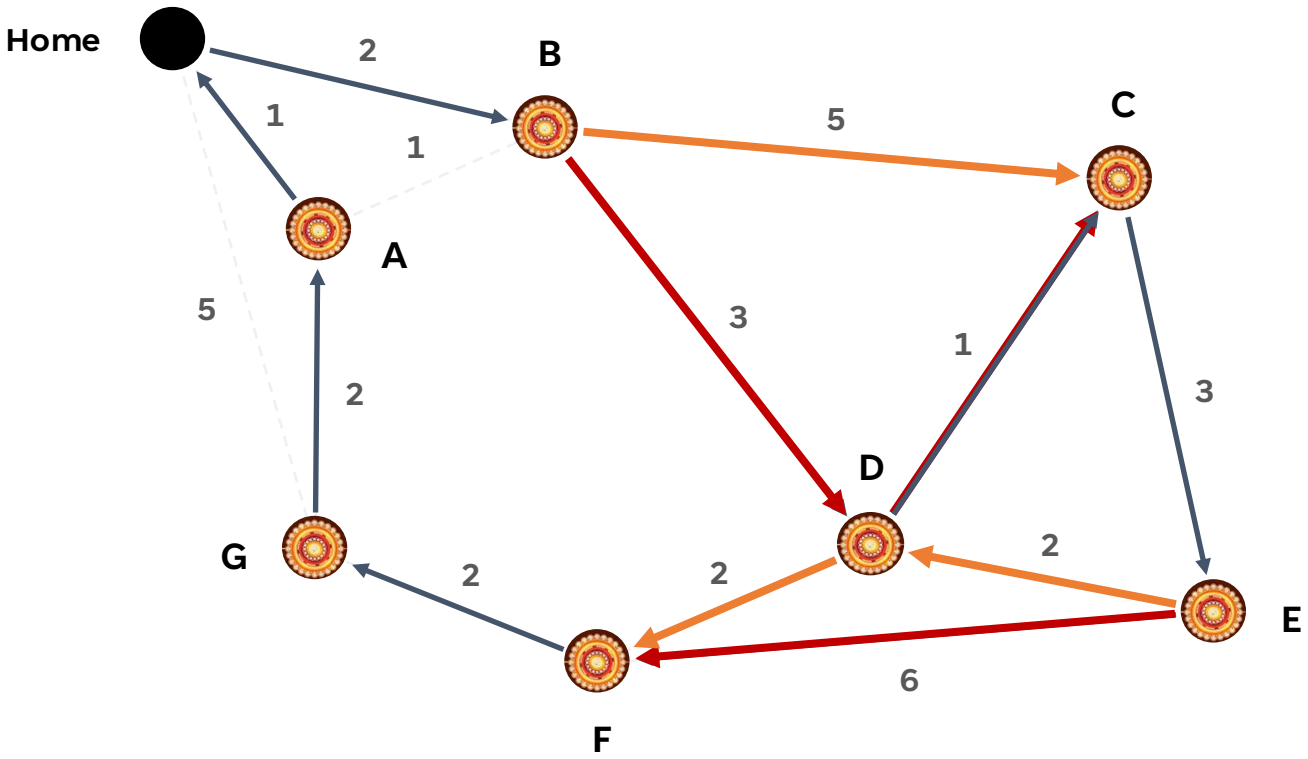
The catch

From inside a valley, a local minimum looks exactly like the global one.

Ways out: restart elsewhere, change the neighbourhood, or accept a worse move now and then.

Next: our Rakhi route gets stuck at 20 km, while the best is 19 km.

ESCAPE 1: CHANGE THE NEIGHBOURHOOD



Route length: **19 km** (optimum)

New move: relocate. Take one cousin out of the route and visit them at a different point.

Same rule: accept only a shorter route.

Step	Move	Route	km
Start	Stuck for swaps	H-B-D-C-E-F-G-A-H	**20**
Iter 1	Move D to after E	H-B-C-E-D-F-G-A-H	**19**

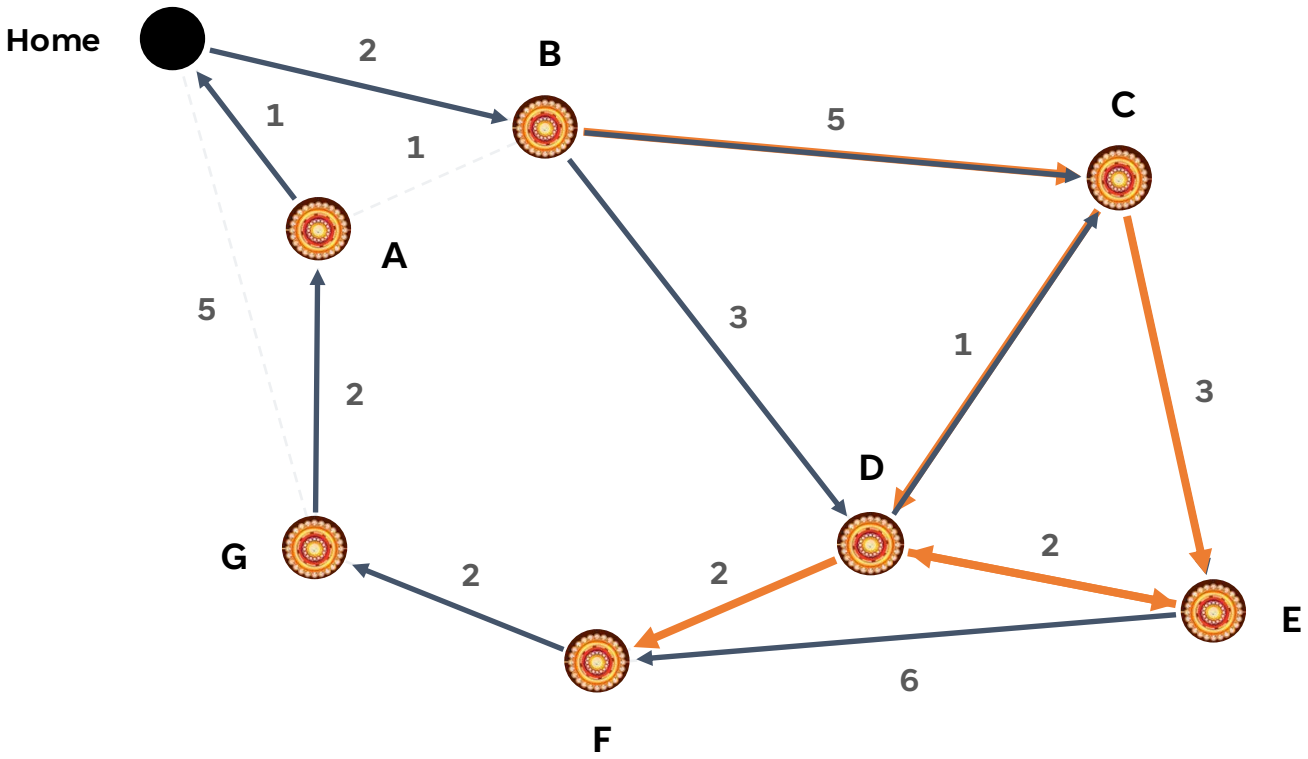
① **Take D out: drop B→D, D→C and E→F**

② **Visit D after E: add B→C, E→D and D→F**

The long E→F road (6 km) is gone: 19 km, the **global minimum**.

A dead end for one kind of move need not be a dead end for another.

ESCAPE 2: TABU SEARCH



Route length: 19 km optimum, but accepted

Same swap move, new rule: take the best move that is not tabu, even if the route gets longer.

Tabu list: recent moves that may not be undone.

Plain hill climbing stops here: the best swap (C and D) gives 21 km, which is worse.

Step	Move	Route	km	Tabu
Start	Stuck for swaps	H-B-D-C-E-F-G-A-H	**20**	-
Iter 1	Swap C and D (worse, accepted)	H-B-C-D-E-F-G-A-H	**21**	**C-D**
Iter 2	Swap D and E (C-D blocked)	H-B-C-E-D-F-G-A-H	**19**	**D-E**

Tabu search accepts one worse step, cannot go back, and reaches the **19 km optimum**.